



UNIVERSITY OF MORATUWA
Faculty of Information Technology

B.Sc. Honours in Information Technology
B.Sc. Honours in Information Technology & Management
Level 2 – Semester 1 Examination
IN 2100 – OBJECT ORIENTED PROGRAMMING

Time Allowed: 3 hours

December 2022

ADDITIONAL MATERIAL

None

INSTRUCTIONS TO CANDIDATES

1. This paper contains five (5) questions on five (5) pages including the cover page.
2. The total marks obtainable for this examination is 100. The marks assigned for each question are included in square brackets.
3. This examination accounts for 50% of the module assessment.
4. This is a closed book examination.
5. Answer **ALL** questions.

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Question 1 [Total Marks: 20 marks]

Write a Java program by following the instructions given below. If you make any assumptions while implementing the Java program, state them appropriately as Java Comments.

- a) Create an interface named **IUniversity** and add an abstract method named **allocateFunds**. This method has a String parameter, and the name of the parameter is **nameFaculty**. Moreover, the return type of the **allocateFunds** method should be *void*.
[5 marks]
- b) Create a class named **AbUniversity**, and it should implement **IUniversity**. The **AbUniversity** class should have a *private* variable named **numStudents**. This variable is to store the number of students at AbUniversity.
[2 marks]
- c) Add a constructor to the **AbUniversity** class. You should initialize the **numStudents** variable inside this constructor.
[3 marks]
- d) Override the **allocateFunds** method in the **AbUniversity** class. Inside this method, write codes to display a message as “Done”.
[5 marks]
- e) Add a method named **checkNumberOfStudents** inside the **AbUniversity** class. Inside this method, write code to check the number of students at AbUniversity. If the number of students is greater than 50,000, then display a message as “A-Grade university”.
[5 marks]

Question 2 [Total Marks: 20 marks]

- a) What is the difference between checked and unchecked exceptions in Java?
[6 marks]
- b) List down two (2) differences between abstract classes and interfaces in Java?
[2 x 2 marks]
- c) Write a suitable Java program to show the occurrence of an unchecked exception. The execution of your program should lead to an unchecked exception.
[10 marks]

Continued...

Question 3 [Total Marks: 20 marks]

Consider the following Java program (Figure 1) with the line numbers for answering Question 3 (a), (b), and (c).

```

1  public class Tester{
2
3      public static void main(String[] args) {
4
5          ThreadTester tTester = new ThreadTester();
6          Thread tOne = new Thread(tTester);
7          tOne.start();
8      }
9  }
10
11 class ThreadTester implements Runnable {
12
13     public void run() {
14
15         for (int j=0; j < 50; j += 3) {
16             System.out.println(j);
17             if (j*j-41 == 400) {
18                 break;
19             }
20             try {
21                 Thread.sleep(5000);
22             } catch (InterruptedException e) {
23                 throw new RuntimeException(e);
24             }
25         }
26     }
27 }

```

Figure 1: Thread testing (Tester.java)

a) What is the output of this program?

[8 marks]

b) Explain the functionality of the statement in Line 21.

[6 marks]

c) Assume that the method `run()` is called in Line 7, instead of calling `start()`. Now Line 7 looks like:

`tOne.run();`

Explain how the execution of the program changes after this code modification.

[6 marks]

Continued...

Question 4 [Total Marks: 20 marks]

Consider the following Java program (Figure 2) with the line numbers for answering Question 4 (a) and (b).

```

1  public class Starter {
2
3      static final int MAX = 5;
4
5      public static void main(String[] args) {
6          int[] numArray = new int[MAX];
7          int count = MAX-1;
8
9          for(int i=0; i < numArray.length - (MAX/2); i++) {
10             numArray[i] = i;
11         }
12
13         while (count > 1) {
14             System.out.println(numArray[count]);
15             count--;
16         }
17     }
18 }

```

Figure 2: Array testing (Starter.java)

a) What is the output of this Java program?

[8 marks]

b) Explain the use of the **static** and **final** keywords in Line 3.

[5 marks]

c) Describe the concept of immutability in Java String with the aid of a coding example.

[7 marks]

Question 5 [Total Marks: 20 marks]

Assume that you are assigned to develop a software to manage inventories of crops such as fruits, vegetables, etc. that are collected from farmers. For this development, you should use object-oriented programming principles and Java programming language. If you make any assumptions, state them clearly.

a) List three (3) classes that you propose to implement.

[3 x 2 marks]

Continued...

- b) List at least three (3) relevant instance variables with their data types for each class.
[3 x 3 marks]
- c) Add a static variable to two (2) of the classes that you mentioned in Question 5 Part (a).
[2 x 2 marks]
- d) Add a constant to one (1) of the classes that you mentioned in Question 5 Part (a).
[1 mark]

End of Paper.